AI ACADEMY · LEVEL 7 · GRADE 12 · WEEK 20 · STUDENT LESSON

Privacy and data governance

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

Before you start

Recall Research ethics and consent. Explain why a recorded result needs its input conditions and evidence source before it can support a claim.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Explain the mechanism and terminology of privacy and data governance. Try the recall before opening notes.

Your learning target

I can explain the **mechanism** and terminology of privacy and **data governance**.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Research ethics and consent. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Sensitivity analysis. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Statistical tests and assumptions. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

This week you will practise:

Explain the **mechanism** and terminology of privacy and **data governance**

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

Recall Research ethics and consent. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

> **My starting explanation**

This week's end goal

Explain privacy and **data governance** using the records and boundaries of this fictional investigation.

Notice the evidence

> Investigate the supplied Privacy and data governance case. Separate observations, given facts and assumptions before choosing an explanation.

Data governance specifies who may use which records, for what purpose and for how long. Coded data can still be reidentifiable.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

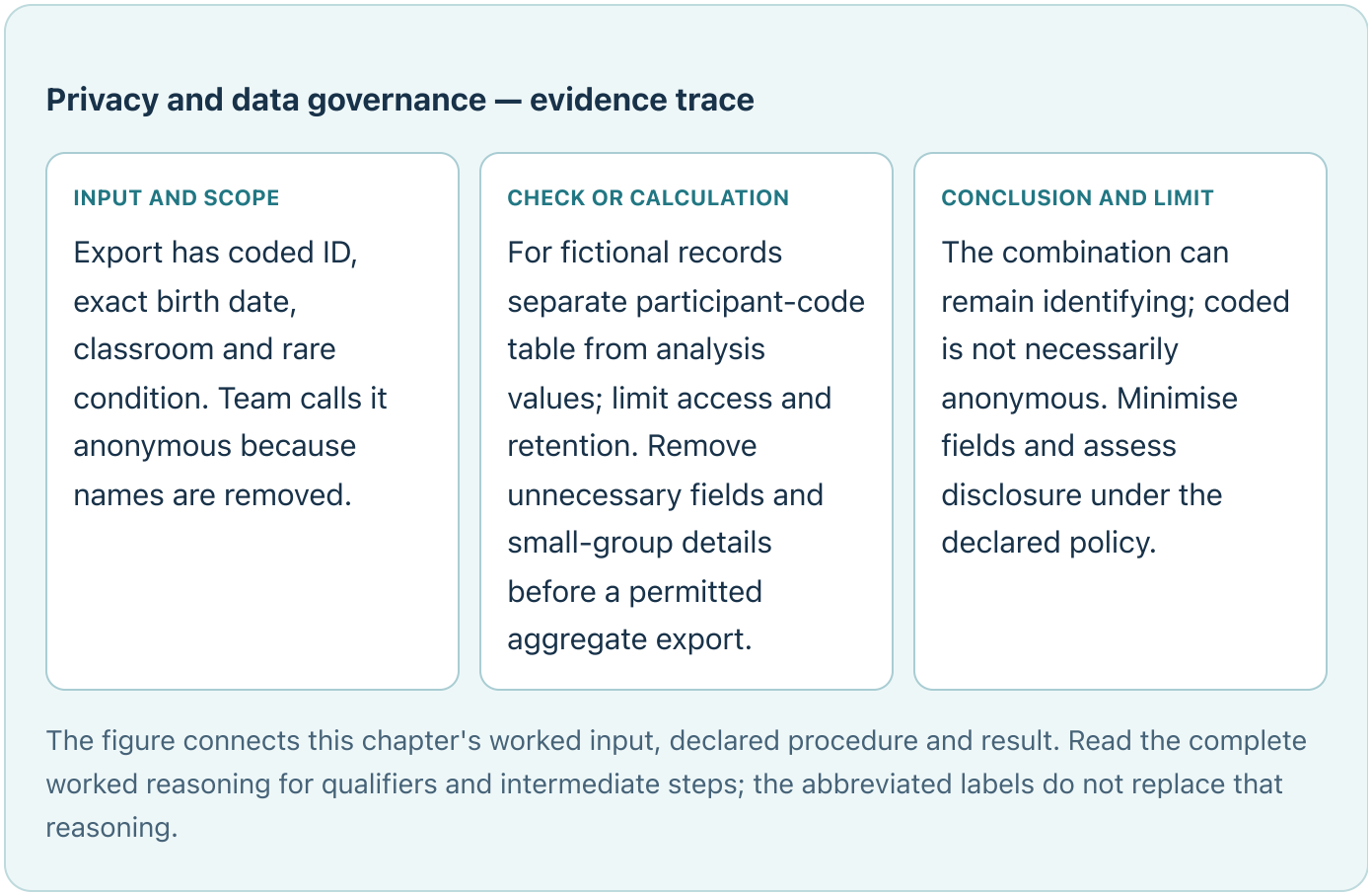
> **List the given facts and one fact that is not supplied.**

STUDENT LESSON · 3 OF 18

See how it works

> Follow the mechanism for Privacy and data governance. Connect each labelled input, deciding step and result.

For fictional records separate participant-code table from analysis values; limit access and retention. Remove unnecessary fields and small-group details before a permitted aggregate export.



Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

> Compare the supplied cases for Privacy and data governance. Identify the detail that changes the conclusion.

Compare a familiar case with a changed case

Export has coded ID, exact birth date, classroom and rare condition. Team calls it anonymous because names are removed. Show the intermediate steps and give the conclusion with its scope.

Keep the procedure fixed. Explain what changed in the inputs and result.

Change one thing in the pictured case

Change one thing

Change: Publish the identity key with the analysis table.

Predicted result

Prediction: The records become directly linkable to names and breach this access design.

Why the result follows

Reason: Separation only helps while the linkage key remains restricted.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

Use the vocabulary of Privacy and data governance to explain a concrete decision, rather than only memorising definitions.

Word	Meaning
Data governance	Rules for purpose, access, quality and lifetime
Pseudonym	A code replacing direct identification
Disclosure risk	The possibility of learning identifying or sensitive facts

Use two terms to explain a step in this week's example.

STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Privacy and data governance, predict a step, then compare with its explanation.

Export has coded ID, exact birth date, classroom and rare condition. Team calls it anonymous because names are removed.

The combination can remain identifying; coded is not necessarily anonymous. Minimise fields and assess disclosure under the declared policy.

Worked explanation · Privacy and data governance

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

> The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Privacy and data governance

Govern data by purpose and access. A fictional project needs task scores and random participant codes, but not names.

The facts we will use

> **Supplied facts**
>
> The analysis table holds codes and scores. A separate restricted key links codes to names only for authorised withdrawal handling.

AI ACADEMY / AI-7/20

Privacy and data governance

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Separate tables

Keep analysis records apart from the identity key.



2. Limit access

Analysts receive scores and codes, not the name linkage.



3. Define retention

Apply the approved deletion schedule to both stores and listed copies.

CHANGE ONE THING

Publish the identity key with the analysis table.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Separate tables	Keep analysis records apart from the identity key.
2. Limit access	Analysts receive scores and codes, not the name linkage.
3. Define retention	Apply the approved deletion schedule to both stores and listed copies.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Pseudonymous codes reduce routine exposure but are not the same as irreversible anonymity. Governance specifies purpose, readers, locations and retention.

What this example does not establish

Keep the stated limit

Small groups or unusual records may still be identifiable without names; evaluate re-identification risk.

Keep the stated limit

Read the labelled picture: Point to the analysis table and the restricted identity key. Why are the codes **pseudonym**ous rather than guaranteed anonymous, and what changes if the key is published?

Keep the stated limit

Change one thing: Publish the identity key with the analysis table. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Privacy and data governance claim and explain a repair using the supplied evidence.

A size threshold or coding scheme alone does not guarantee privacy.

Check the claim against the facts

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Locate the earliest unsupported step

A report omits this qualification: A size threshold or coding scheme alone does not guarantee privacy. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

State one repair and the checks you would rerun.

STUDENT LESSON · 8 OF 18

Practise together

> Practise Privacy and data governance with support, then explain the deciding step in your own words.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Explain your prediction to a partner before comparing results.

Practise with less help: Trace the worked case

Use workbook W2 for the connected example "Privacy and **data governance**". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Privacy and **data governance**. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

> Point to the analysis table and the restricted identity key. Why are the codes **pseudonym**ous rather than guaranteed anonymous, and what changes if the key is published?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Privacy and data governance. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

Predict, investigate and preserve the evidence

Export has coded ID, exact birth date, classroom and rare condition. Team calls it anonymous because names are removed. Show the intermediate steps and give the conclusion with its scope.

Record	Your evidence
Given input and fixed rule	
Prediction before checking	

Record	Your evidence
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

Build your understanding

> Move from recall to tracing, diagnosis and transfer in Privacy and data governance. Explain why each step is justified.

Explain → apply → challenge

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

A report omits this qualification: A size threshold or coding scheme alone does not guarantee privacy. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

> **Explain which evidence supports your conclusion and which claim remains unproved.**

STUDENT LESSON · 11 OF 18

Connect your project

> Apply this week's idea to Reproducible AI Research Study. Keep an annotated decision and the evidence supporting it.

Add a labelled privacy and **data governance evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of privacy and **data governance**.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Evidence for your project

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

STUDENT LESSON · 12 OF 18

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

> **Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.**

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My limited claim and a useful next check

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Privacy and data governance case for supported practice. Repeated examples do not become fresh assessments.

Complete evidence record

Privacy and data governance — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Export has coded ID, exact birth date, classroom and rare condition. Team calls it anonymous because names are removed.	For fictional records separate participant-code table from analysis values; limit access and retention. Remove unnecessary fields and small-group details before a permitted aggregate export.	The combination can remain identifying; coded is not necessarily anonymous. Minimise fields and assess disclosure under the declared policy.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Apply a new case independently

Aggregate contains one member in a subgroup; policy forbids subgroup reports smaller than5. Decide.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

Aggregate contains one member in a subgroup; policy forbids subgroup reports smaller than5. Decide.

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Privacy and data governance, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Govern data by purpose and access. A fictional project needs task scores and random participant codes, but not names.

By the end, I can	How I will show it
Explain the mechanism and terminology of privacy and data governance	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Privacy and data governance without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of privacy and **data governance**

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building

> block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Govern data by purpose and access. A fictional project needs task scores and random participant codes, but not names.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Apply the approved deletion schedule to both stores and listed copies.

1. Define retention

2. Separate tables

3. Limit access

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Define retention Apply the approved deletion schedule to both stores and listed copies.

Pseudonymous codes reduce routine exposure but are not the same as irreversible anonymity. Governance specifies purpose, readers, locations and retention.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Publish the identity key with the analysis table. Which conclusion is supported by the supplied facts?

1. The records become directly linkable to names and breach this access design.

2. We can decide the result without examining the changed **input** or the procedure.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The records become directly linkable to names and breach this access design. Separation only helps while the linkage key remains restricted.

Repair a missing building block

Review Research ethics and consent (ai-7/19). Goal: Explain the **mechanism** and terminology of research ethics and consent. Short practice: Identify which uses the drawing consent permits and excludes.

Can the unrelated project reuse the drawings under this agreement, and what new authorisation would be needed?. Return to this week and explain what you changed.

Review Sensitivity analysis (ai-7/17). Goal: Explain the **mechanism** and terminology of sensitivity analysis. Short practice: Classify 0.4, 0.6 and 0.8 at cutoffs 0.5, 0.7 and exactly 0.6.

Which equality rule decides the boundary, and can these unlabelled scores identify the best cutoff?. Return to this week and explain what you changed.

Review Statistical tests and assumptions (ai-7/14). Goal: Explain the **mechanism** and terminology of statistical tests and assumptions. Short practice: Calculate each paired difference and the observed sum.

Count null arrangements with absolute sum at least four, compute p, and explain the decision at threshold 0.05.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

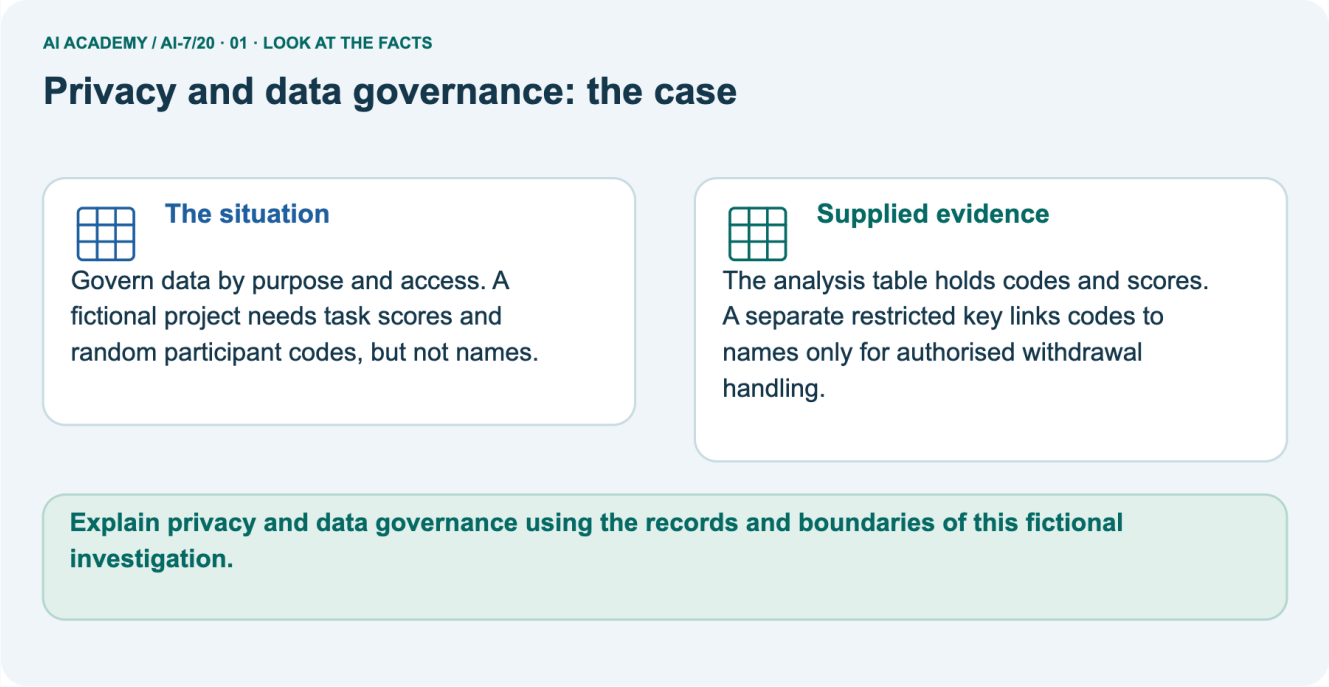
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Privacy and data governance graphic using its labels, deciding evidence and relevant limitation.

01. Privacy and data governance: the case



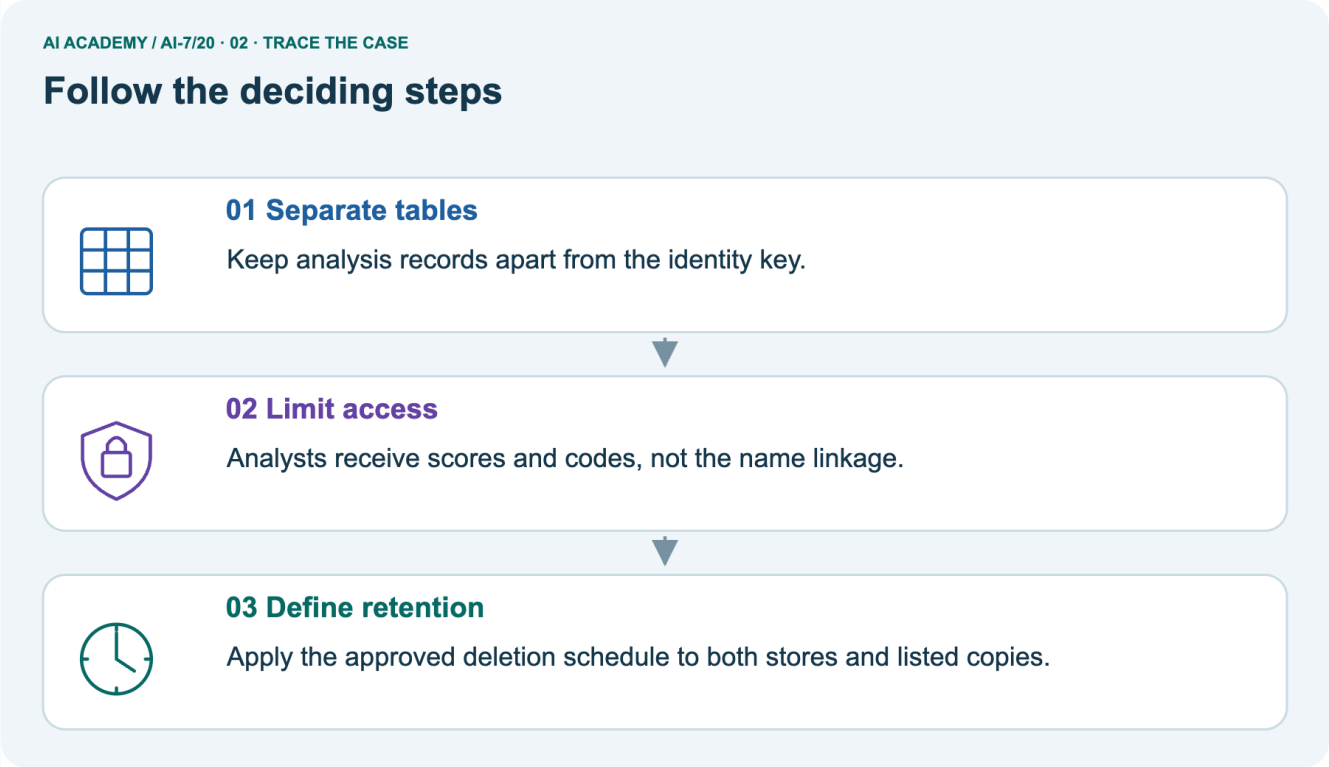
Follow the labelled picture with the matching explanation. Open the vector diagram

Govern data by purpose and access. A fictional project needs task scores and random participant codes, but not names. Supplied facts: The analysis table holds codes and scores.

A separate restricted key links codes to names only for authorised withdrawal handling.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

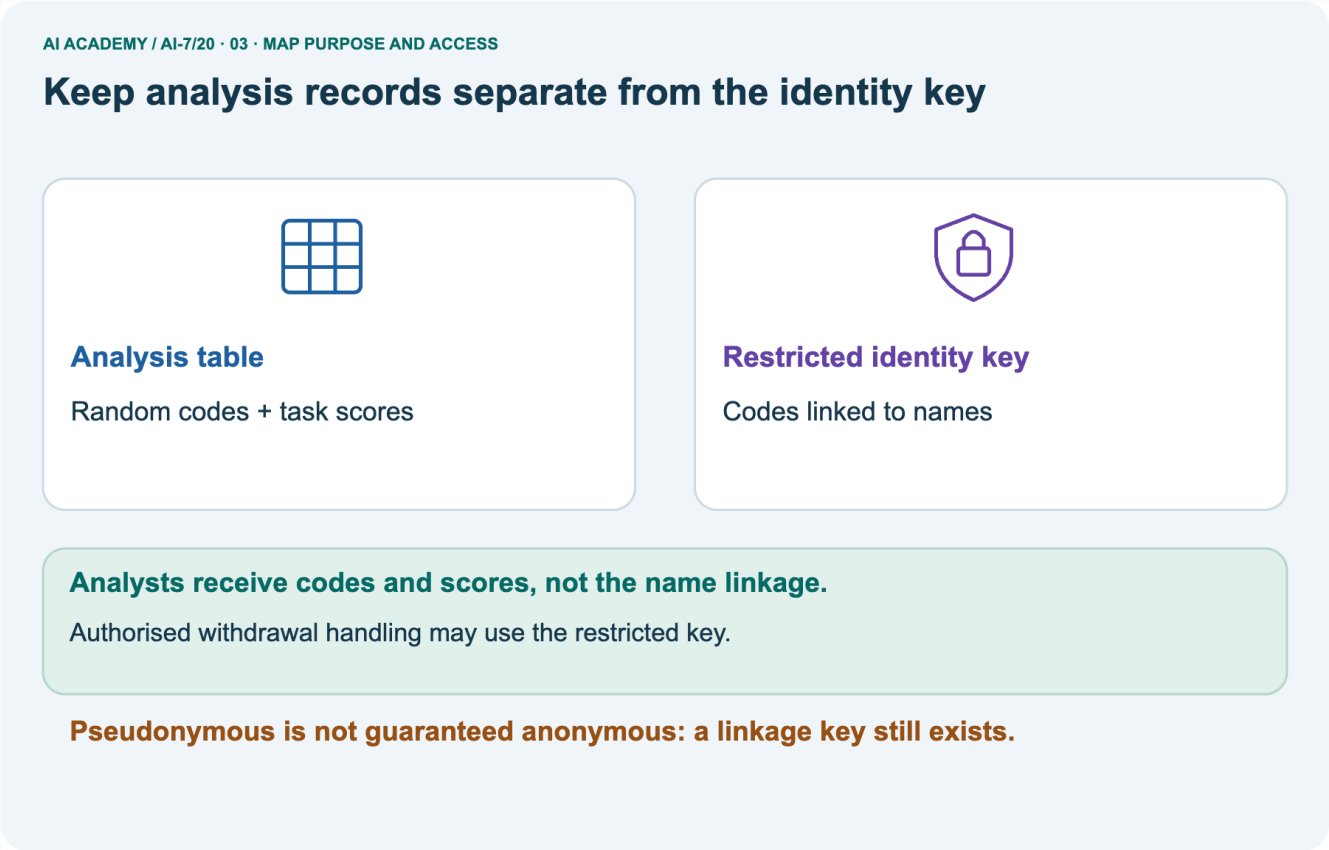


Follow the labelled picture with the matching explanation. Open the vector diagram

Separate tables: Keep analysis records apart from the identity key. → Limit access: Analysts receive scores and codes, not the name linkage. → Define retention: Apply the approved deletion schedule to both stores and listed copies.

Point to each numbered stage and explain which recorded **input** it uses.

03. Keep analysis records separate from the identity key

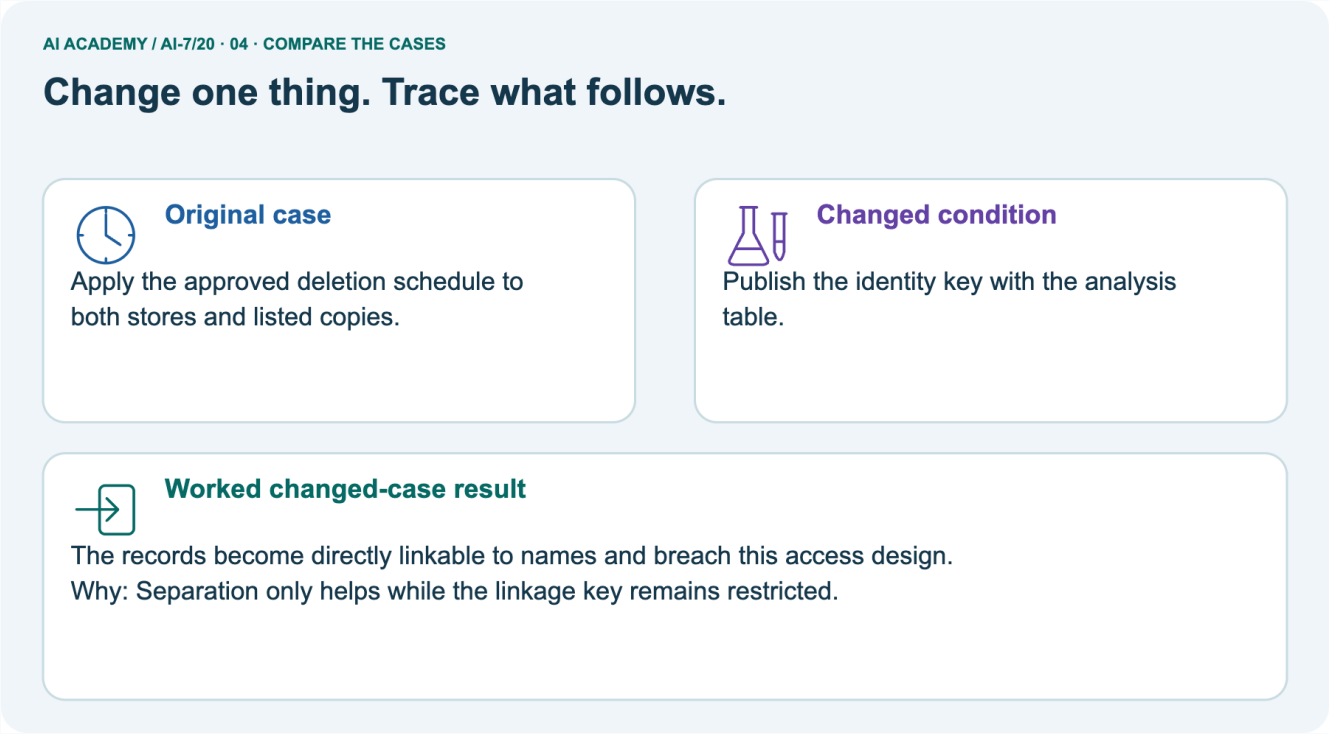


Follow the labelled picture with the matching explanation. Open the vector diagram

Analysts receive random participant codes and scores. A separate restricted key links codes to names only for authorised withdrawal handling. Keeping the key restricted reduces exposure; it does not guarantee anonymity.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



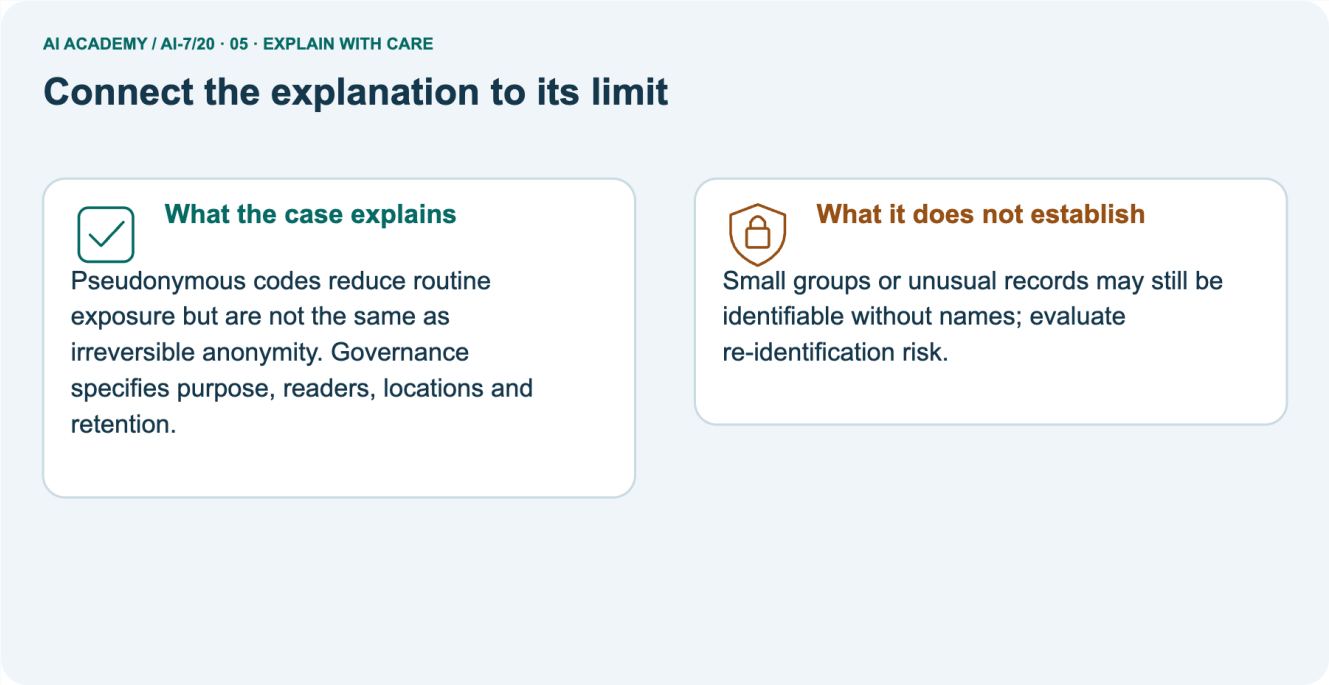
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Apply the approved deletion schedule to both stores and listed copies. Changed condition: Publish the identity key with the analysis table.

Expected result: The records become directly linkable to names and breach this access design. Reason: Separation only helps while the linkage key remains restricted.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



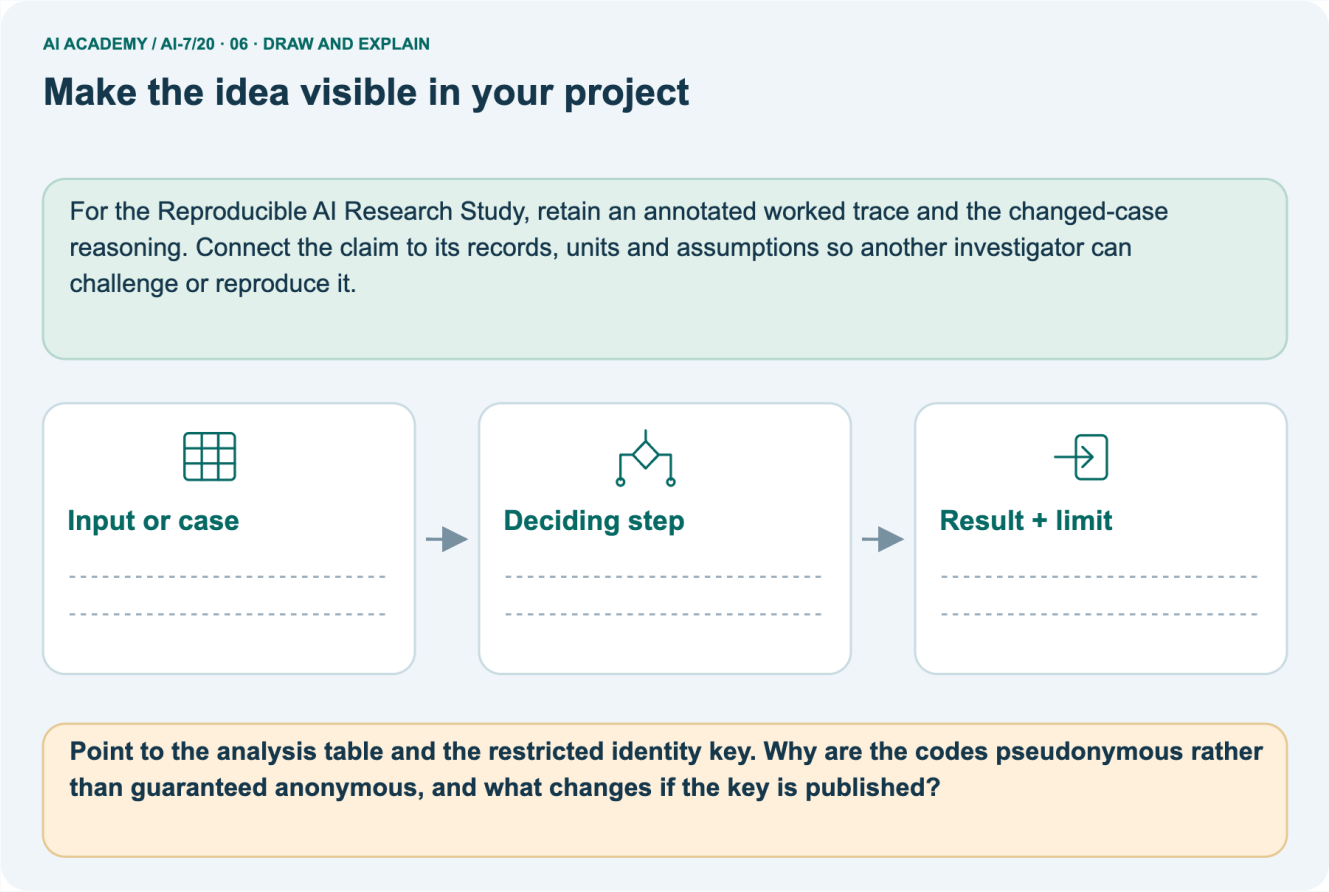
Follow the labelled picture with the matching explanation. Open the vector diagram

Pseudonymous codes reduce routine exposure but are not the same as irreversible anonymity. Governance specifies purpose, readers, locations and retention.

Limit: Small groups or unusual records may still be identifiable without names; evaluate re-identification risk.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Point to the analysis table and the restricted identity key.

Why are the codes **pseudonym**ous rather than guaranteed anonymous, and what changes if the key is published?

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.